

Assignment 2: Informed Search

1. (15 points) Suppose that we are given an admissible heuristic function h . Consider the following function

$$h'(n) = \begin{cases} h(n), & \text{if } n \text{ is the initial state } s, \\ \max\{h(n), h'(n') - c(n', n)\}, & \text{where } n' \text{ is the predecessor node of } n, \end{cases}$$

where

$$c(n', n) := \min_a c(n', a, n).$$

Prove that h' is consistent.

Solution:

From the problem, we know that the definition of h' , for non-initial node n with predecessor n' ,

$$h'(n) = \max\{h(n), h'(n') - c(n', n)\}$$

Since the maximum of this two is at least as large as each of them, we have:

$$h'(n) \geq h'(n') - c(n', n)$$

$$h'(n') \leq h'(n) + c(n', n)$$

Since

$$c(n', n) = \min_a c(n', a, n) \leq c(n', a, n)$$

for every action a , we have

$$h'(n') \leq h'(n) + c(n', n) \leq h'(n) + c(n', a, n).$$

Therefore,

$$h'(n') \leq c(n', a, n) + h'(n),$$

which is exactly the definition of consistent. Therefore, h' is consistent.

2. (25 points) Consider a search problem over a graph $G = (N, E, C)$, where N are nodes, E are directed edges between nodes, and the weight of an edge $e \in E$ is given by $c(e)$, where $c(e) > 1$, for all $e \in E$. We assume that there is a unique goal state g .
- (a) (5 points) Consider the heuristic h_1 that counts the number of edges between a given state s and the goal state g . Show that h_1 is consistent.
 - (b) (4 points) Find a consistent heuristic h_2 that dominates the heuristic h_1 . The heuristic h_2 cannot be equal to h_1 , nor can it be the optimal heuristic h^* that computes the true cost of reaching g from the state s .
 - (c) (8 points) Suppose that we add edges between states in the graph G , but keep the heuristic h_1 unchanged. Is h_1 still admissible and consistent? Prove this claim or give a counterexample.
 - (d) (8 points) Suppose that we remove edges from the graph G , but keep the heuristic h_1 unchanged. Is the heuristic still admissible and consistent? Prove this claim or give a counterexample.

Solution:

a)

Assume $h_1(s)$ is the min number of edges from s to the goal state g .

Let's we take a successor of s , which is t , then we have: $1 + h_1(t)$.

Since $h_1(s)$ is the min number of edges from s to g , we have:

$$h_1(s) \leq 1 + h_1(t)$$

Because every edge cost $c(e) \geq 1$,

$$h_1(s) \leq 1 + h_1(t) \leq c(s, t) + h_1(t)$$

Hence h_1 is consistent.

b)

$$c_{min} = \min_{e \in E} c(e)$$

$$c_{min} > 1$$

Choose a constant α

$$1 < \alpha < c_{min}$$

$$h_2(s) = \alpha h_1(s) \text{ and } h_2(s) \geq h_1(s)$$

Let's we take a successor of s , which is t , then we have: $1 + h_1(t)$.

Since $h_1(s)$ is the min number of edges from s to g , we have:

$$h_1(s) \leq 1 + h_1(t)$$

$$h_2(s) \leq \alpha + h_2(t)$$

Since $\alpha \leq c(s, t)$,

$$h_2(s) \leq c(s, t) + h_2(t)$$

Hence h_2 is consistent.

c) No

Counterexample:

Let we assume a graph:

$$A \rightarrow B \rightarrow g$$

and each edge cost 2.

$$h_1(A) = 2, h_1(B) = 1, h_1(g) = 0$$

Now, we add one more edge $e(A, g)$, and we set the cost $c(A, g) = 1.5$.

Let we check admissible:

$$h^*(A) = 1.5$$

$$h_1(A) = 2 > 1.5 = h^*(A)$$

h_1 is not admissible.

Let we check consistent:

$$h_1(A) \leq c(A, g) + h_1(g)$$

$$2 \leq 1.5 + 0$$

h_1 is not consistent.

d) Still Admissible and consistent

Let we check admissible:

Removing edges will not produce a shorter path; it can only make the optimal cost larger or unchanged. Assume the optimal cost of old graph is $h_{old}^*(s)$, the optimal cost of new graph is $h_{new}^*(s)$, we have:

$$h_{new}^*(s) \geq h_{old}^*(s)$$

$$h_1(s) \leq h_{old}^*(s) \leq h_{new}^*(s)$$

h_1 is admissible.

Let we check consistent:

New graph edge set: E_{new}

Old graph edge set: E_{old}

$$E_{new} \subseteq E_{old}$$

$$h_1(s) \leq c(s, t) + h_1(t) \quad \forall (s, t) \in E_{old}$$

Since every edge in the new graph is also an edge in the old graph, we have:

$$h_1(s) \leq c(s, t) + h_1(t) \quad \forall (s, t) \in E_{new}$$

h_1 is consistent.

3. (20 points) Answer the following questions.

- (a) (10 points) Prove that if h is consistent, then it must be admissible (remember that for any goal state t , $h(t) = 0$).
- (b) (10 points) Suppose that h, h' are heuristics such that h dominates h' . Show that if h is admissible, then so is h' ; does the same claim hold for consistent heuristics? I.e. if h is consistent and dominates a heuristic h' , is h' consistent as well? Prove or provide a counterexample.

Solution:

a)

Let n be any node, and let

$$n = n_0 \rightarrow n_1 \rightarrow \cdots \rightarrow n_k = t$$

be an optimal path from n to a goal node t . Since h is consistent, for every edge (n_i, n_{i+1}) ,

$$h(n_i) \leq c(n_i, n_{i+1}) + h(n_{i+1})$$

$$h(n_0) \leq c(n_0, n_1) + c(n_1, n_2) + \cdots + c(n_{k-1}, n_k) + h(n_k)$$

$$h(n_0) \leq c(n_0, n_1) + c(n_1, n_2) + \cdots + c(n_{k-1}, n_k)$$

The right-hand side is the optimal cost from n to a goal node, which is $h^*(n)$. Therefore,

$$h(n) \leq h^*(n)$$

Hence h is admissible.

b)

1.

Since h dominates h' ,

$$h'(n) \leq h(n)$$

Since h is admissible,

$$h(n) \leq h^*(n)$$

Therefore,

$$h'(n) \leq h(n) \leq h^*(n)$$

Therefore h' is admissible.

2.

No

Counterexample:

$$A \rightarrow B \rightarrow G, \quad c(A, B) = 1, \quad c(B, G) = 1$$

Let G be the goal node.

Define h

$$h(G) = 0, \quad h(B) = 1, \quad h(A) = 2$$

Then h is consistent:

$$h(A) = 2 \leq 1 + h(B) = 2$$

$$h(B) = 1 \leq 1 + h(G) = 1$$

Now define h'

$$h'(G) = 0, \quad h'(B) = 0, \quad h'(A) = 2$$

$$h'(n) \leq h(n)$$

h dominates h' .

However,

$$h'(A) \leq c(A, B) + h'(B)$$

$$2 \not\leq 1 + 0$$

Therefore, h' is not consistent.

4. (20 points) You have learned before that A^* using graph search is optimal if $h(n)$ is consistent. Does this optimality still hold if $h(n)$ is admissible but inconsistent? Using the graph in Figure 1, let us now show that A^* using graph search returns a sub-optimal solution path (S, B, G) from start node S to goal node G with an admissible but inconsistent $h(n)$. We assume that $h(G) = 0$. Give nonnegative integer values for $h(A)$ and $h(B)$ such that A^* using graph search returns the non-optimal solution path (S, B, G) from S to G with an admissible but inconsistent $h(n)$, and tie-breaking is not needed in A^* .

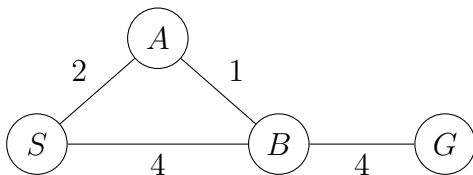


Figure 1: Graph for heuristic search.

Solution:

a)

Let us take:

$$\begin{aligned}
 h(A) &= 5, & h(B) &= 0, & h(G) &= 0. \\
 h^*(G) &= 0, & h^*(B) &= 4, & h^*(A) &= 1 + 4 = 5.
 \end{aligned}$$

Thus,

$$\begin{aligned}
 h(A) &= 5 \leq h^*(A) = 5, \\
 h(B) &= 0 \leq h^*(B) = 4, \\
 h(G) &= 0 \leq h^*(G) = 0.
 \end{aligned}$$

Therefore, h is admissible. However, h is not consistent. For edge $A \rightarrow B$,

$$\begin{aligned}
 h(A) &\leq c(A, B) + h(B) \\
 5 &\not\leq 1 + 0,
 \end{aligned}$$

b)

Step 1:

Expand S ,

$$\begin{aligned}
 g(A) &= 2, & f(A) &= g(A) + h(A) = 2 + 5 = 7, \\
 g(B) &= 4, & f(B) &= g(B) + h(B) = 4 + 0 = 4. \\
 f(B) &< f(A),
 \end{aligned}$$

A^* expands B first and generates G :

$$g(G) = 4 + 4 = 8, \quad f(G) = 8.$$

Step 2:

Expand A , B is already closed.

Then, pop G

Therefore, A^* returns

$$(S, B, G)$$

But the optimal path is

$$(S, A, B, G)$$

Thus, A^* using graph search returns a sub-optimal solution path (S, B, G) from start node S to goal node G with an admissible but inconsistent $h(n)$.

5. (20 points) Consider the minimax search tree below (Figure 2). In the figure, black nodes are controlled by the MAX player, and white nodes are controlled by the MIN player. Payments (terminal nodes) are squares; the number within denotes the amount that the MIN player pays to the MAX player (an amount of 0 means that MIN pays nothing to MAX). Naturally, MAX wants to maximize the amount they receive, and MIN wants to minimize the amount they pay. Suppose that we use the α - β pruning algorithm. Consider the minimax tree given in Figure 2.
- (a) (10 points) Assume that we iterate over nodes from **right to left**; what are the arcs that are pruned by α - β pruning, if any?
- (b) (10 points) Does your answer change if we iterate over nodes from **left to right**?

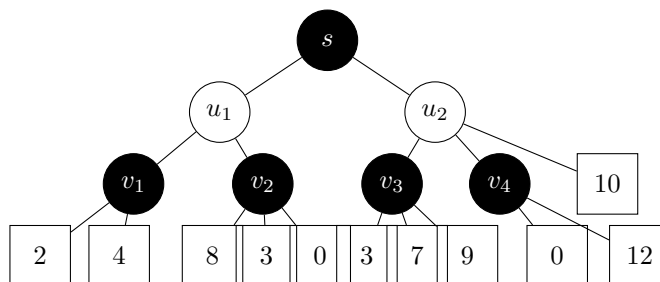


Figure 2: A minimax search tree.

Solution:

Minimax value: 9

a)

Pruned:

$$v_4 \rightarrow 0$$

$$u_1 \rightarrow v_1$$

b)

Pruned:

$$v_2 \rightarrow 3$$

$$v_2 \rightarrow 0$$

6. **(Optional, bonus points)** In this question, we will formally prove that the minimax algorithm described in class produces an equilibrium. An extensive form game is defined by a set of nodes V and a set of directed edges E , defining a tree. The root of the tree is the node r . Let $V_{\max}$ be the set of nodes controlled by the MAX player and $V_{\min}$ be the set of nodes controlled by the MIN player. We often refer to the MAX player as player 1, and to the MIN player as player 2. Furthermore, let $\text{desc}(v)$ be the descendants of a node v in the search tree. A strategy for the MAX player is a mapping $s_1 : V_{\max} \rightarrow V$; similarly, a strategy for the MIN player is a mapping $s_2 : V_{\min} \rightarrow V$. In both cases, $s_i(v) \in \text{chld}(v)$ is the choice of child node that will be taken at node v . We let S_1, S_2 be the set of strategies for the MAX and MIN player, respectively.

The leaves of the minimax tree are payoff nodes. There is a payoff $a(v)$ associated with each payoff node v , where $a(v)$ is the amount that the MAX player receives and $-a(v)$ is the amount that the MIN player receives if the node v is reached. The MAX player wants to maximize their payoff, and the MIN player wants to minimize the amount that they pay. More formally, the utility of the MAX player from v is $u_{\max}(v) = a(v)$ and the utility of the MIN player is $u_{\min}(v) = -a(v)$. The utility of a player from a pair of strategies $s_1 \in S_1, s_2 \in S_2$ is simply the utility they receive by the leaf node reached when the strategy pair (s_1, s_2) is played.

Definition 1 (Nash Equilibrium). A pair of strategies $s_1^* \in S_1, s_2^* \in S_2$ for the MAX player and the MIN player, respectively, is a Nash equilibrium if no player can get a strictly higher utility by switching their strategy. In other words:

$$\forall s \in S_1 : u_1(s_1^*, s_2^*) \geq u_1(s, s_2^*);$$

$$\forall s' \in S_2 : u_2(s_1^*, s_2^*) \geq u_2(s_1^*, s').$$

Definition 2 (Subgame). Given an extensive form game $\langle V, E, r, V_{\max}, V_{\min}, \vec{a} \rangle$, a subgame is a subtree of the original game, defined by some arbitrary node v set to be the root node, and all of its descendants (i.e. its children, its children's children etc.), denoted by $\text{desc}(v)$. Leaf nodes and node control are the same as in the original extensive form game.

Definition 3 (Subgame-Perfect Nash Equilibrium (SPNE)). A pair of strategies $s_1^* \in S_1, s_2^* \in S_2$ is a sub-perfect Nash equilibrium if it is a Nash equilibrium for any subtree of the original game tree.

- (a) Prove that the minimax algorithm shown in class outputs a subgame-perfect Nash equilibrium.
- (b) Consider a node v in an adversarial search problem. Assume that v is controlled by the MAX player (Player 1). Prove that α - β pruning only removes subtrees of v that offer Player 1 a payoff that is weakly lower than their payoff in the subtree rooted in v under a subgame-perfect Nash equilibrium.